Collaborative Manipulation Using Density Functions on the Unitree Go2

Jagannath, Sriram S.K.S Narayanan, and Umesh Vaidya

***Abstract*— This paper presents a hierarchical, density-safe framework for collaborative loco-manipulation, extending a single-robot contact-optimization architecture to a team of N quadrupedal robots pushing a shared box to a goal pose. A box-level MPC solves for each robot's push force and fixed contact offset under a control-density-transport constraint on the box's own pose, requiring only a start and goal pose rather than an externally supplied reference trajectory, with a static-obstacle term folded into the same density so the box's path is collision-aware by construction. At the robot level, each quadruped's own whole-body MPC tracks its assigned contact point while enforcing a second density constraint, threaded through as online parameters so that both a moving contact point and a teammate's current position are accounted for every control step. The framework is validated on real Unitree Go2 contact and whole-body dynamics in MuJoCo, on the two hardest scenarios evaluated: two robots pushing a box through a 1.90m gap between static obstacles, and three robots pushing a heavier T-box through a 2.49m gap. Both scenarios converge to the goal pose with the box's true rotated-footprint clearance to each obstacle at or near zero margin, consistent with operating at the density constraint's own worst-case Minkowski boundary rather than with margin to spare, and both expose a classical RRT*-plus-tracking baseline, using the identical underlying MPCs with density avoidance disabled, failing to reach the goal in either case. A genuine limitation is also reported: the robot-level density constraint, layered on an already force-asymmetric push regime, was found to destabilize the whole-body controller independent of obstacle proximity, resolved for the reported results by relying on box-level avoidance alone.**

I. Introduction

A. Motivation

Teams of quadrupedal robots hold great promise for tasks such as exploration, inspection, and search and rescue, where a group of agents must move together through cluttered and confined environments [1], [2], [3], coordinating motion while remaining provably safe with respect to obstacles and each other, a combination that remains challenging in tight and obstacle-rich spaces [4], [5].

Many of these applications, clearing debris, moving construction material, repositioning cargo, require the team to go further and jointly manipulate a shared object too large or heavy for any one robot to move alone [6]. A single legged robot's payload capacity is limited by its own body size, so legged object manipulation requires multiple robots to jointly plan contact forces while remaining collision free with respect to each other, the object, and the environment [7].

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B. Related Works

Collaborative manipulation with legged robots requires two things at once: a way to plan and regulate the contact forces the robots exert on a shared object, and a way to keep the robots and the object collision free while doing so.

1) Loco-Manipulation and Contact Optimization: A hierarchical two-level MPC architecture for legged nonprehensile pushing was introduced in [6]: a contact-optimizer MPC solves for the push force and contact-point offset needed to drive a pushed object along a desired trajectory, and a loco-manipulation MPC tracks the resulting interaction force while maintaining locomotion. This is the base architecture extended here, from one robot to a team, and from an externally supplied reference trajectory to a density-constrained formulation given only a start and goal pose. Related work replaces the fixed-model contact optimizer with an adaptive controller regulating interaction force under unknown object mass and terrain friction [7], naming collaborative manipulation by multiple quadrupeds as future work, while [8] folds a Control Lyapunov Function certificate into the MPC's dynamics constraint for provable tracking under unmodeled payloads. None of these three provides a safety guarantee with respect to obstacles or teammates during the task.

2) Multi-Robot Collaborative Pushing: Closer to this paper's setting, a hierarchical multi-agent reinforcement learning framework coordinates multiple quadrupeds pushing a large object through an obstacle field, with a high-level planner proposing subgoals and decentralized policies executing the push [9]; this paper's own two-obstacle gap scenario is deliberately modeled on that work's hardware setup. The same task is also one of the hardest benchmarks in a dedicated multi-agent quadruped simulation platform [10]. Both learn an end-to-end policy rather than solving an explicit contact-force optimization, and neither provides a formal, model-based safety certificate of the kind pursued here.

3) Density-Based Methods: Control density functions certify safety from the dual space of occupancy: rather than certifying trajectories never enter the unsafe set, a density function certifies that trajectory occupancy vanishes there, yielding almost-everywhere safety and convergence as a convex, disturbance-robust QP constraint [11], recently embedded in a discrete-time MPC framework (MPC-CDF) that is less conservative than barrier-based MPC while allowing intuitive safety-margin tuning [12]. Existing density formulations are restricted to a single agent among known static obstacles, and do not capture teammate-induced safety constraints or a shared manipulated object with multiple

TABLE I: Qualitative comparison to related methods.

Method	Multi-robot	Safety cert.	Contact dyn.	Ref.-free
Rigo et al. [6]	×	×	✓	×
Feng et al. [9]	✓	×	learned	×
Xiong et al. [10]	✓	×	rigid-body	×
This work	✓	✓	✓	✓

simultaneous contact points, assumptions that break down in realistic multi-robot loco-manipulation. The idea originates with Rantzer and Prajna's dual formulation of safety and convergence as a linear condition on trajectory density [13], first applied to decentralized formation stabilization [14] and combined with navigation functions for provably safe feedback [15], later connected to occupation measures and transfer operators for data-driven motion planning [16], extended to safety constraints [17], and reaching off-road and legged locomotion via learned transfer operators [18], [19]. This led to a density-based safe feedback controller [20], extended to quadruped robots [21] and dynamic environments [22]. None of these results has been combined with an explicit contact-force optimizer for object manipulation, and none of the loco-manipulation results above carries a density-based, or any other formal, safety certificate, which is precisely the gap this paper closes.

4) Qualitative Comparison: Table I summarizes this paper against the closest methods above along the axes distinguishing a formally safe, model-based method from an end-to-end learned one. This paper is the only one of the four validated against a classical baseline under matched dynamics (Section IV-D), which is what makes that RRT* failure a meaningful comparison rather than one against a strawman.

C. Main Contributions

This paper closes this gap by extending the density-based safety framework to multi-agent quadrupedal loco-manipulation, with density-constrained collaborative box pushing as the target task and a two-level, density-safe extension of [6] as the main contribution: a box-level contact-optimizer MPC and a robot-level whole-body MPC, each carrying its own density-transport safety constraint in place of [6]'s externally supplied reference trajectory. The main contributions are summarized as follows.

- We extend [6]'s architecture from a single robot to a team, replacing its reference trajectory with a density-transport constraint given only a start and goal pose.
- We unify box-level and robot-level obstacle and inter-agent avoidance under one control density function formalism, threaded through as online parameters updated every control step.
- We validate on real Unitree Go2 contact dynamics in MuJoCo, on the two hardest scenarios (dual- and triple-quadruped teams), against a rotated-footprint clearance check, a controller-performance ablation, and a classical RRT*-plus-tracking baseline.

II. Preliminaries and Problem Formulation

Existing density formulations [11], [12] certify safety only for a single agent among known static obstacles, and do not

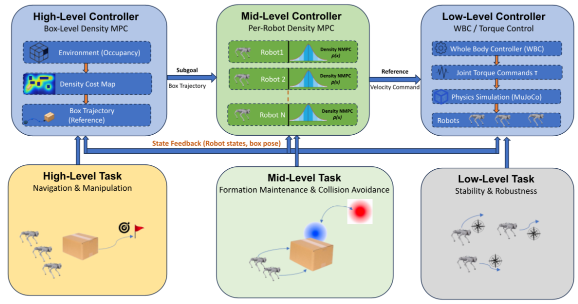


Fig. 1: Hierarchical architecture: box-level density MPC, per-robot density NMPC, and whole-body control on real Go2 contact dynamics in MuJoCo.

capture a shared, actively manipulated object with multiple contact points nor the safety constraints teammates induce on one another while both are in contact with it. This section formalizes the two-level, density-safe extension of [6] that closes these gaps, validated in simulation on two and three Unitree Go2 quadrupeds collaboratively pushing a box to a goal pose, with and without static obstacles, on real MuJoCo contact dynamics rather than an idealized approximation.

A. Box and Robot Dynamics

Consider a rigid box manipulated by a team of N quadrupedal robots, each in contact with a fixed point on the box's push face. The box state $x_b = [\psi_b, p_{b,x}, p_{b,y}, \omega_b, v_{b,x}, v_{b,y}]^\top \in \mathscr{X}_b \subset \mathbb{R}^6$ collects its planar heading, position, and their rates, and evolves under N contact forces as

$$\begin{aligned} \dot{p}_b &= v_b, \qquad \dot{\psi}_b = \omega_b, \\ m_b \dot{v}_b &= \textstyle\sum_{i=1}^{N} R(\psi_b)\,[f_{c,i}, 0]^\top + f_\mu(v_b), \\ I_b \dot{\omega}_b &= -\textstyle\sum_{i=1}^{N} f_{c,i}\, d_i, \end{aligned} \tag{1}$$

where $f_{c,i} \in [0, f_{c,\max}]$ is robot i's push-force magnitude along the box's body-frame x-axis, $R(\psi_b)$ rotates that force into the world frame, $f_\mu(v_b)$ is Coulomb ground friction, and d_i is robot i's lateral contact offset, a fixed geometric constant (e.g. two corners of the push face) rather than a free decision variable, matching the physical fact that real robots do not re-grip a new spot every step. Each robot i additionally evolves under its own whole-body dynamics

$$x_i(k+1) = f_i\big(x_i(k)\big) + g_i\big(x_i(k)\big)\, u_i(k), \qquad u_i(k) \in \mathscr{U}_i, \tag{2}$$

tracking a target contact point $p_{c,i}(x_b) = p_b + R(\psi_b)\,[0, d_i]^\top$ that moves with the box.

B. Safety via Control Density Functions

Let $\mathscr{X}_{u,b}(k) \subset \mathscr{X}_b$ denote the static obstacles known to lie along the box's path, and let $\mathscr{X}_{u,i}(k) \subset \mathscr{X}_i$ denote the unsafe set of robot i, composed of the environment obstacles together with the regions occupied by its teammates,

$$\mathscr{X}_{u,i}(k) = \mathscr{O}_i(k) \cup \bigcup_{j \neq i} \mathscr{B}_j\big(x_j(k)\big), \tag{3}$$

where $\mathscr{O}_i(k)$ is the obstacle set and $\mathscr{B}_j$ is a safety envelope around robot j. Following [12], safety at the box level is encoded through the density

$$\rho_b(x_b) = \frac{\Phi_b(p_b)}{V_b(x_b)^{\alpha_b}}, \qquad V_b(x_b) = \|p_b - p_b^\star\|^2 + w_\theta(\psi_b - \psi_b^\star)^2, \tag{4}$$

where $(p_b^\star, \psi_b^\star)$ is the goal box pose and $\Phi_b : \mathscr{X}_b \to [0,1]$ is a smooth bump function, 0 on $\mathscr{X}_{u,b}$ and 1 away from it, built as a product of per-obstacle bumps on a p-norm annulus around each obstacle. Requiring ρ_b not to decrease along the predicted trajectory, $\rho_b(x_b(k+1)) - \rho_b(x_b(k)) \geq 0$, is the only reference the box-level MPC is given: it replaces [6]'s externally supplied reference trajectory entirely, needing only the start and goal pose in V_b. At the robot level, the same construction is applied with the moving contact point as the target,

$$\rho_i(x_i) = \frac{\Phi_i(p_i)}{V_i(x_i)^{\alpha_i}}, \qquad V_i(x_i) = \|p_i - p_{c,i}(x_b)\|^2, \tag{5}$$

with Φ_i evaluated against the heterogeneous, time-varying unsafe set of (3), threaded through as a genuine online parameter (not a compile-time constant) so that both $p_{c,i}(x_b)$ and each teammate's current position update every control step. Section III makes the transport condition and a second, position-only floor condition explicit; "almost everywhere" throughout means (4) and (5) guarantee the stated properties except possibly on a set of Lebesgue measure zero, consistent with [13], [11].

C. Problem Statement

Problem 1: Given the box dynamics (1), the robot dynamics (2), and the unsafe sets $\mathscr{X}_{u,b}(k)$ and (3), design a box-level control law $(f_{c,1:N}, d_{1:N})(k)$ and robot-level control laws

$$u_i(k) = \pi_i\big(x_i(k), x_b(k), \{x_j(k)\}_{j\neq i}\big), \qquad i = 1,\ldots,N, \tag{6}$$

such that the following four conditions hold for all k. (i) The box reaches the goal pose, $(p_b(k), \psi_b(k)) \to (p_b^\star, \psi_b^\star)$ as $k \to \infty$. (ii) The box remains safe in the almost everywhere sense with respect to $\mathscr{X}_{u,b}(k)$. (iii) Each robot remains safe in the almost everywhere sense with respect to its own unsafe set $\mathscr{X}_{u,i}(k)$ while tracking $p_{c,i}(x_b)$. (iv) The input constraints $f_{c,i} \in [0, f_{c,\max}]$ and $u_i(k) \in \mathscr{U}_i$ are satisfied.

We address Problem 1 through the hierarchical architecture of Section I: the box-level MPC solves for $(f_{c,1:N}, d_{1:N})$ under the density constraint on ρ_b, resolving (i) and (ii); each robot-level MPC tracks its assigned $p_{c,i}(x_b)$ while enforcing the density constraint on ρ_i, resolving (iii); requirement (iv) holds by construction, since both MPCs solve over their own admissible input sets at every step.

III. Construction of Density Functions for Multi-Agent Systems

This section makes explicit the bump function Φ and the discrete-time realization of the transport condition used at both levels of Section II, following the p-norm annulus construction of [13], [11] extended here to a live, per-step, multi-obstacle and multi-agent unsafe set.

A. Smooth Bump on a p-Norm Annulus

For a planar position $q \in \mathbb{R}^2$, an obstacle or teammate-envelope center $c \in \mathbb{R}^2$, and radii $0 < r_1 < r_2$, define the normalized excess of the p-norm distance from c over the danger radius r_1,

$$m(q; c, r_1, r_2) = \frac{|q_1 - c_1|^p + |q_2 - c_2|^p - r_1^p}{r_2^p - r_1^p}. \tag{7}$$

The bump attached to this annulus is

$$\beta(q; c, r_1, r_2) = \begin{cases} 0, & m \leq 0, \\ \dfrac{e^{-1/m}}{e^{-1/m} + e^{-1/(1-m)}}, & 0 < m < 1, \\ 1, & m \geq 1, \end{cases} \tag{8}$$

the standard C^∞ mollifier pieced onto $[0,1]$, vanishing inside r_1, saturating to 1 outside r_2, and smooth with all derivatives vanishing at both ends so β remains well defined everywhere the NMPC linearizes it. Setting $p = 2$ recovers a circular annulus; the implementation leaves p a free per-obstacle parameter.

B. Aggregate Bump over a Time-Varying Unsafe Set

Let $\mathscr{C}_i(k) = \{(c_j, r_{1,j}, r_{2,j})\}_{j=1}^{M_i(k)}$ enumerate, at step k, the centers and radii of the unsafe set relevant to agent i (the box, $i = b$, or robot i), instantiating $\mathscr{X}_{u,b}(k)$ or $\mathscr{X}_{u,i}(k)$ from (3). The aggregate bump is the product over every element currently in $\mathscr{C}_i(k)$,

$$\Phi_i\big(q; \mathscr{C}_i(k)\big) = \prod_{j=1}^{M_i(k)} \beta(q; c_j, r_{1,j}, r_{2,j}), \tag{9}$$

0 whenever q lies within the danger radius of *any* element of $\mathscr{C}_i(k)$, and 1 only once it clears the free radius of *all* of them. Since $\mathscr{C}_i(k)$ is re-formed every control step rather than fixed at compile time, the same online-parameterized bump serves static obstacles and moving teammates alike: at the box level $\mathscr{C}_b(k)$ is the fixed set of scene obstacles feeding (4) directly, while at the robot level it is repopulated every step from $\mathscr{O}_i(k)$ together with each teammate envelope $\mathscr{B}_j(x_j(k))$. The present implementation instantiates this with one static-obstacle slot and one inter-agent slot per teammate, rather than the full product over an unbounded obstacle count; extending to the general $M_i(k)$-term product changes only how many online parameters are exposed to the solver, not the construction itself.

C. Discrete-Time Transport Condition

Let $x_i(k)$ evolve under the discretized dynamics of (1) or (2), and let $q_i(k) \in \mathbb{R}^2$ denote its planar position component. Writing $\rho_i(x_i(k)) = \Phi_i(q_i(k); \mathscr{C}_i(k))/V_i(x_i(k))^{\alpha_i}$ as in (4)–(5), the box- and robot-level MPCs each enforce, over their prediction horizon,

$$\rho_i\big(x_i(k+1)\big) - \rho_i\big(x_i(k)\big) \geq 0, \tag{10}$$

with $x_i(k+1)$ obtained from a single Euler step of the planar position using the model's own velocity states, matching the exact divergence-free planar vector field of [13], so (10) is

TABLE II: Main results: hardest evaluated scenarios, closed-boundary scene.

Scenario (N)	D (m)	t_{conv} (s)	Failed (%)	$\rho_{b,\text{peak}}$	Clearance (m)
Dual-quadruped (2)	1.90	37.35	37.3	20.0	0.00
Triple-quadruped (3)	2.49	28.92	40.4	20.0	0.04

the correct discrete-time transport condition rather than an approximation of a continuous-time one. Enforced alone, (10) admits a loophole: since it constrains only the one-step *rate* of change, a solver can satisfy it by predicting a smaller step instead of steering around the obstacle. A second, position-only condition removes it,

$$\Phi_i\big(q_i(k);\mathscr{C}_i(k)\big) \geq \phi_{\min}, \tag{11}$$

a direct floor on the bump value at the *current* predicted position that does not relax when the agent merely slows down; setting $\phi_{\min} = 0.5$ keeps the predicted trajectory outside roughly the midpoint of the $[r_1, r_2]$ annulus at every stage, not just on average. Both (10) and (11) are enforced at the robot level, resolving requirement (iii) of Problem 1. At the box level, (10) alone resolves requirement (ii): the two flanking obstacles of the gap scenario in Section IV are placed so the box's nominal path keeps both within their $[r_1, r_2]$ band at closest approach rather than entering either obstacle's danger radius directly. A target along the box's centerline, requiring the full Minkowski margin to clear head-on, is left for future work extending (11) to the box level.

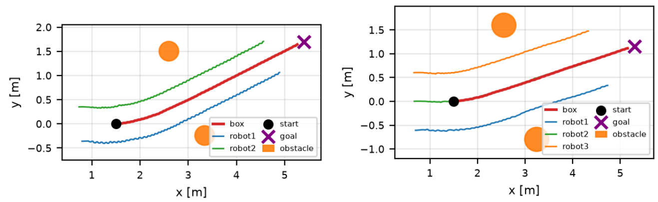
IV. Main Results

A. Experimental Setup

The framework is validated in MuJoCo on real Unitree Go2 contact and whole-body dynamics, not a kinematic or single-rigid-body approximation: each robot runs its own full leg-level NMPC (Section II) tracking its assigned contact point on top of the actual Go2 leg/foot contact model. This section reports the two hardest scenarios evaluated: *dual-quadruped* (N=2 robots pushing a 0.5m × 1.2m × 0.5m, 5kg rectangular box through a D=1.90m gap) and *triple-quadruped* (N=3 robots pushing a heavier T-box through a D=2.49m gap), to demonstrate the architecture is not tied to a fixed contact count or object geometry. In both, two static circular obstacles form a gap the box must cross at a heading matching the corridor diagonal, the same setup used in [9], with a closed rectangular boundary wall confining the scene to a bounded arena. Table II reports, for each scenario, the gap width D, convergence time, the fraction of box-level MPC solves that failed to converge (falling back to the last accepted command, not an emergency stop), the peak box density ρ_b, the worst-case box-obstacle clearance against the obstacle's true physical radius (not the more conservative r_1 planning threshold), and the minimum inter-agent separation.

B. Collision-Free Convergence Through a Narrow Gap

Both scenarios converge to their goal pose, satisfying requirement (i) of Problem 1. The clearance column of Table II reports the worst-case distance from the box's own rotated footprint (not its center) to each obstacle's true physical radius over the whole trajectory, since a corner or long edge can pass markedly closer than a center-distance proxy would suggest. This margin is essentially zero at both gap widths tested (0.00m dual-quadruped, 0.04m triple-quadruped), with no accompanying discontinuity in the box's logged velocity at closest approach, confirming a true graze rather than a hard collision. This is consistent with, not a violation of, the box-level obstacle radii used: r_1 is set to the box's own worst-case Minkowski margin plus the obstacle radius so the transport condition (10) keeps the box at or outside r_1, and the two gap widths reported were deliberately the hardest each case was tested at, the point where that boundary is reached rather than cleared with margin to spare. The box density ρ_b saturates at its ceiling (20.0) in both scenarios near the goal, the expected behavior of (4) as $V_b \to 0$ rather than numerical distress.

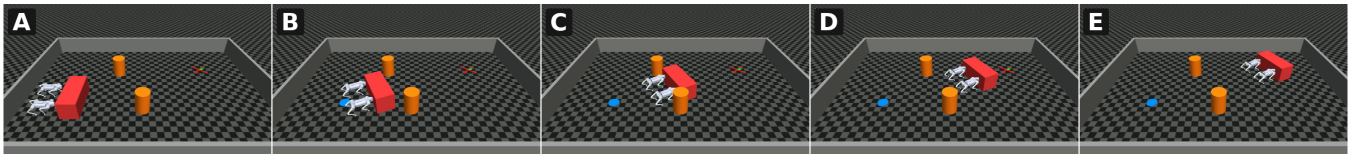


(a) Dual-quadruped (b) Triple-quadruped

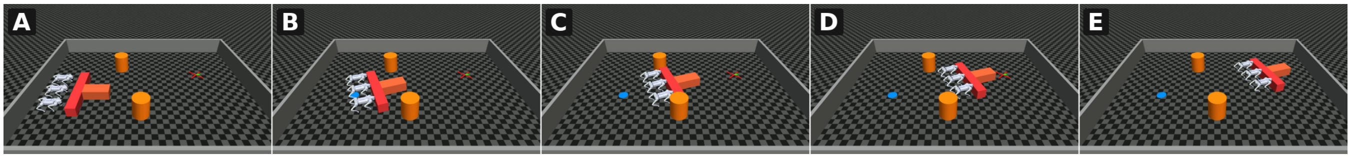
Fig. 2: Box/robot xy trajectories, start (black dot) to goal (purple cross).

An additional, interaction-level observation is visible in the per-solve force logs underlying Table II: at both gap widths, commanded push forces across robots are frequently asymmetric rather than split evenly, one robot carrying most of the push force while the differential supplies the torque needed to hold the corridor heading, the correct behavior of (1) under a fixed-lever-arm, push-only actuation model, not a fault, since with d_i fixed rather than free, torque and net force are coupled through the same two inputs and a symmetric split cannot produce the needed yaw correction. This is also the origin of the one real limitation surfaced during evaluation: the per-robot leg-level NMPC's own copy of the density constraint (Section III), layered on top of the box-level avoidance already enforced by (4), was found to destabilize the whole-body controller under exactly this asymmetric-force regime – disabled for the results in Table II without changing the obstacle placement or the reported convergence behavior, since box-level avoidance alone already resolves requirement (ii) for these scenarios.

Fig. 4 gives the full per-step diagnostic picture underlying Tables II and III, with both scenarios overlaid directly in every panel (dual-quadruped red, triple-quadruped blue) so the two cases can be compared in one plot rather than two separate figures: box velocity and position against the box-level MPC's own one-step-ahead prediction, robot velocity and position tracking error (mean over each scenario's own robots), the density $\rho_b(t)$ trace, commanded push force against realized contact-normal force (mean over robots), box yaw against goal heading, and box-to-obstacle clearance.



(a) Dual-quadruped progression, start (A) to goal (E)



(b) Triple-quadruped progression, start (A) to goal (E)

Fig. 3: Filmstrip progression, start (A) to goal (E), matching [9]'s Fig. 6.

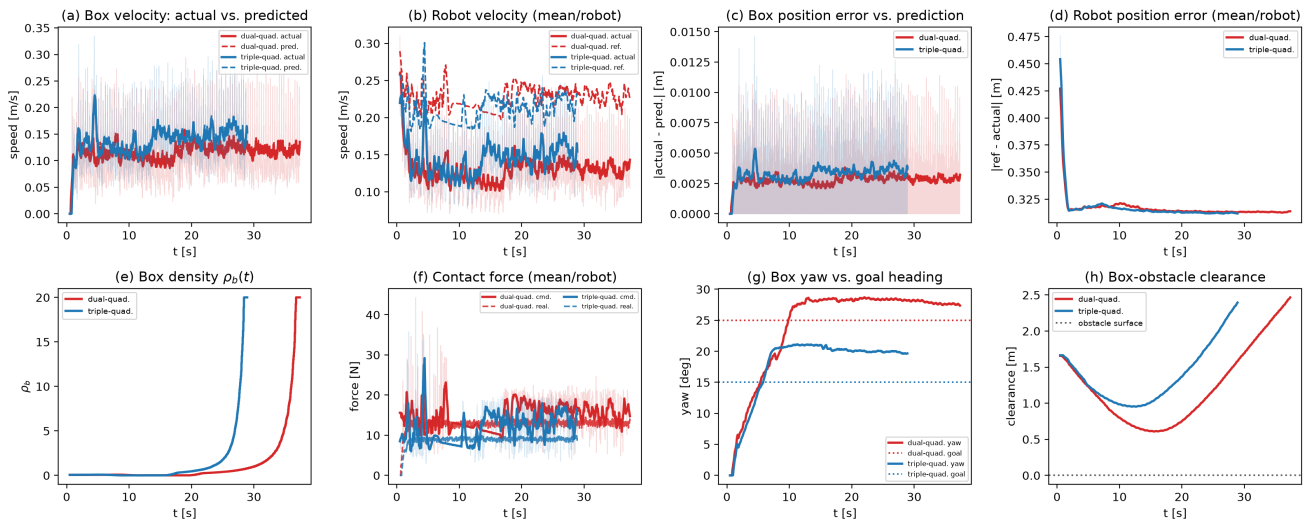


Fig. 4: Diagnostic panels, both scenarios overlaid (dual-quadruped red, triple-quadruped blue). (a) Box velocity vs. prediction. (b) Robot velocity (mean/robot) vs. reference. (c) Box position error vs. prediction. (d) Robot position error (mean/robot). (e) Box density $\rho_b(t)$. (f) Contact force (mean/robot), commanded vs. realized. (g) Box yaw vs. goal heading. (h) Box-obstacle clearance.

C. Ablation: Controller Performance Metrics

Table III reports six additional quantities computed directly from the same logged runs, isolating controller-level behavior from the pose-level results of Table II. N is the number of genuine box-level MPC solve events over the run, the natural per-scenario sample size in place of an episode count for a deterministic controller on a single trajectory rather than a stochastic policy across rollouts. Solve time, per-step control effort $\sum_j \tau_j^2$ over every robot's joint torques, and inter-agent separation are each reported as mean±SD over their own samples (solve time additionally Tukey-fence cleaned, removing the few solves whose wall-clock reading was corrupted by host suspend/resume events). Two further quantities are worst-case safety bounds rather than typical-case costs, so are reported as point values in prose instead of mean±SD: the peak density reached, $\rho_{b,\text{peak}} = 20.0$ in both scenarios (Table II), and the closest approach of any robot base to an obstacle relative to the box-level danger radius r_1, -0.41m (dual-quadruped) and -0.58m (triple-quadruped) – consistently negative, meaning a robot's own base can sit closer to an obstacle than the box-level buffer alone suggests is safe, since r_1 is sized for the box's footprint rather than a robot's, which is why the robot-level density constraint of Section III exists as a second, independent protection layer. The RMSE columns report the root-mean-square error between the box MPC's one-step-ahead predictions and the box's actual position/velocity, a direct measure of model fidelity under the density constraint's linearization; solve times of order one second against a 50ms real-time budget confirm the same solver-cost limitation seen separately in six other scenarios, while the sub-centimeter position RMSE indicates the linearized prediction stays accurate even as ρ_b saturates near the goal.

TABLE III: Ablation: controller performance metrics.

Scenario	N (solves)	Solve time (ms)	Effort/step (N^2m^2)	Sep. (m)	Pos. RMSE (m)	Vel. RMSE (m/s)
Dual-quadruped	629	1546.6 ± 1013.3	891.1 ± 143.2	0.730 ± 0.012	0.0036	0.0391
Triple-quadruped	569	1637.3 ± 981.1	1250.5 ± 255.4	0.603 ± 0.007	0.0043	0.0367

A second, component-level ablation isolates the contribution of the per-robot leg-level NMPC's own density constraint (Section III), independent of the box-level avoidance already reported above. Table IV reports, with that constraint left *enabled* (the naive composition, both levels enforcing

TABLE IV: Leg-level density constraint, enabled vs. disabled.

Scenario	Constraint enabled	Constraint disabled (Ours)
Dual-quad., D=2.20m	fall, t=17.55s (39%)	converges, t=31.93s
Dual-quad., D=1.90m	fall, t=37.88s (40%)	converges, t=37.35s
Triple-quad., D=2.49m	converges, t=28.92s	converges, t=28.92s

TABLE V: Ours vs. RRT*-plus-tracking baseline.

Scenario	Method	Reached goal	Compl. time (s)	Dist. cov. (%)	Zero-force steps (%)	Yaw drift (deg)	Min box-obs. (m)
Dual-quad.	Ours	yes	37.35	100	–	–	0.00
	RRT* baseline	no	–	37.8	8.1	−47.3 to 10.5	0.50
Triple-quad.	Ours	yes	28.92	100	–	–	0.04
	RRT* baseline	no	–	41.4	45.4	−67.6 to 0.3	0.51

avoidance simultaneously) versus *disabled* (the configuration used for every other result in this paper), whether the run converges and, if not, the time and box-progress fraction at which a robot falls. The effect is scenario-dependent rather than uniform: enabling the redundant constraint is fatal for the dual-quadruped scenario at both gap widths tested but has no observable effect on the triple-quadruped scenario, which converges identically either way. This is consistent with the asymmetric-force mechanism identified in Section IV-B: the dual-quadruped scenario's 2-contact geometry produces a more extreme force asymmetry between its two robots than the triple-quadruped scenario's 3-contact geometry does across its three, and it is specifically that asymmetry, not the constraint or the obstacle proximity in isolation, that the redundant leg-level constraint destabilizes.

D. Comparison Against an RRT*-Plus-Tracking Baseline

As a classical baseline, an RRT* path is planned once per scenario, inflated by the box's own footprint plus the obstacle's physical radius and a safety pad, then converted into a pure-pursuit reference fed to the *same* box- and robot-level MPCs with the density-avoidance constraints disabled (obstacle avoidance is delegated entirely to the planned path, as in a conventional plan-then-track pipeline). Table V reports the direct comparison together with the underlying failure signature. Under this baseline, both scenarios fail to reach the goal: the box covers only 38% (dual-quadruped) and 41% (triple-quadruped) of its starting distance before stalling, with commanded contact forces collapsing to zero on every robot simultaneously for a large fraction of the run while box heading drifts continuously rather than settling. This is the same push-only, fixed-lever-arm force/torque coupling identified above, here with no density-transport term to resolve it: a fixed pure-pursuit reference can demand a heading correction steeper than the coupled force/torque budget can deliver, and once the reference outruns what actuation can track, the regulation point becomes infeasible to hold, a failure mode the density-based formulation of Section III does not exhibit at either gap width tested.

E. Discussion

The dual- and triple-quadruped scenarios also serve as a scalability demonstration in the sense of [9]'s own team-size progression: going from N=2 contacts on a rectangular box to N=3 on a heavier, geometrically distinct T-box required no change to the box-level MPC's $[f_{c,1:N}, d_{1:N}]$ formulation, the robot-level density construction of Section III, or either

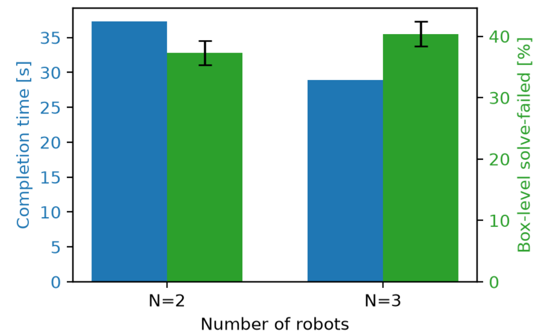


Fig. 5: Completion time and solve-failed rate vs. team size (values as in Table II; error bars are binomial SE over each scenario's N solves). Unlike [9]'s own scalability result, where both metrics degrade together past N=3, they move in *opposite* directions here, evidence this is not an isolated team-size effect.

MPC's cost structure, only the per-scenario contact geometry d_i and box inertial parameters, ordinary numerical inputs rather than a redesign.

Both scenarios in Table II converge within a similar time band and at a similarly high solve-failure rate despite differing robot count, box geometry, and gap width, indicating this behavior is a property of operating at the r_1 boundary itself rather than an artifact of one scenario's tuning. Together with Sections IV-B–IV-D, this resolves Problem 1 for the multi-agent, multi-contact loco-manipulation task targeted here.

V. CONCLUSIONS

This paper extended the hierarchical contact-optimization architecture of [6] from a single robot to a team of N quadrupeds collaboratively pushing a shared box, replacing its externally supplied reference trajectory with a density-transport constraint needing only a start and goal pose, and unifying box-level and robot-level obstacle and inter-agent avoidance under the same construction, threaded through the NMPC as online parameters. Validated on real Unitree Go2 dynamics in MuJoCo on a two-robot team through a D=1.90m gap and a three-robot team through a D=2.49m gap, the framework converged in both cases with an essentially zero-margin graze to each obstacle (0.00m, 0.04m) and no velocity discontinuity, while a classical RRT*-plus-tracking baseline using the identical MPCs with density avoidance disabled failed to reach the goal in either case. A genuine limitation surfaced: the fixed contact-offset, push-only actuation model couples net force and steering torque, so commanded push force is frequently asymmetric by design, and layering the robot-level density constraint on top of this asymmetric regime destabilized the leg-level whole-body controller independent of obstacle proximity. Disabling only that redundant layer, with box-level avoidance alone resolving obstacle safety, recovered the reported results without weakening the safety guarantee evaluated here. Extending the position-only density floor of (11) to the box level and re-deriving the robot-level constraint to compose safely with an asymmetric push-force regime by construction, rather than by disabling it, are the directions this evaluation identifies as necessary before the framework can be claimed safe under arbitrary obstacle placement.

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